

SIH 2026 · PROBLEM STATEMENT 26123

The Reasoning Chain

From a Centralized Robot Fleet to a Working
Decentralized 3-AMR System

Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs)
in Smart Warehouses

■ Removing the Server

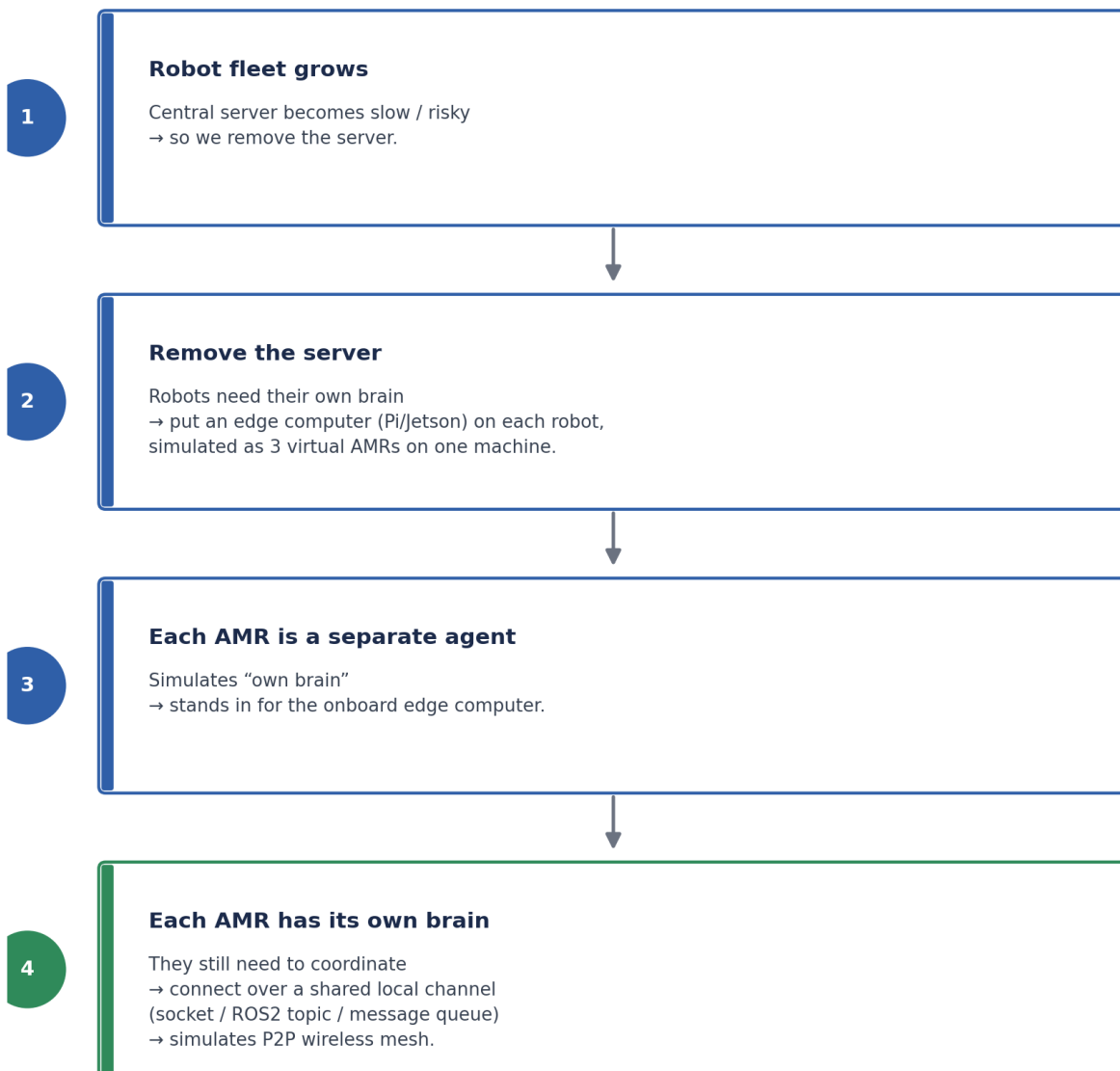
■ Communication & Conflict

■ Blocked Aisles & Re-routing

■ Dashboard

■ Final Proof

Part 1 of 4 — Removing the Central Server



Part 2 of 4 — Communication & Conflict

5

P2P channel exists

Each AMR publishes its position + intended path
→ every other AMR can hear it.



6

All 3 publishing paths

Give them a shared warehouse grid map with narrow aisles (choke points) → set pickups so at least one path overlaps.



7

Paths overlap

Two AMRs claim the same aisle/cell at the same time → conflict happens.



8

Conflict happens

Priority rule decides (e.g. lower battery = right of way) → one proceeds, one yields/reroutes.



9

Rule works, but repeats

Check it doesn't always make the same AMR lose
→ fix it to stay fair and deadlock-free.

Part 3 of 4 — Blocked Aisles & Task Re-routing

10

Simulate a blocked aisle

Manually flag a cell as blocked for one AMR
(say AMR-2) → AMR-2 broadcasts “blocked”.



11

Other AMRs hear “blocked”

They re-plan their own path locally, live,
mid-run.



12

AMR-2's task becomes undoable

Its pickup is broadcast as open → AMR-1 and
AMR-3 “bid” (distance/battery) → closer one
takes over the task.



13

All 3 AMRs broadcasting data

Position / battery / task → build a dashboard
script that just listens to this same channel.

Part 4 of 4 — Dashboard & Final Proof

14

Dashboard listens only

Renders live positions / battery / task status of all 3 AMRs → visual proof.



15

Dashboard must stay passive

It never sends anything back to the AMRs → keeps the system truly decentralized.



16

3-AMR system built and working

Run the same scenario with a dumb stop-and-wait version → time it.



17

Compare both runs

Decentralized version must show $\geq 20\%$ faster completion + zero collisions → final demo proof.